HMR INSTITUTE OF TECHNOLOGY & MANAGEMENT
Hamidpur, Delhi-110036
(An ISO 9001: 2008 certified, AICTE approved & GGSIP University affiliated institute)
E-mail:hmritmdirector@gmail.com, Phone: - 8130643674, 8130643690, 8287461931, 8287453693

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
Weekly Progress Report (WPR)

WPR Dates: 13/2/2026 - 20/2/2026 **WPR of Week:**5

Name: Santosh Kumar

Enrollment Number: 12196502722

Class & Section: CSE-8A

Department: Computer Science and Engineering

Faculty Supervisor's Name: Ms. Indu Sharma

MAJOR PROJECT TITLE: - "AURA : Autonomous User Realization & Assembly"

TARGETS SET FOR THE WEEK:

- Progress the hardware-side prototype using the ESP32 host and low-power node concept.
- Refine node-level indication logic for grouped components and distributed storage.
- Map how value-aware inventory checks should support part sufficiency decisions.
- Document the whole-project highlighting workflow for selected builds.
- Prepare clearer notes on how the host bridges smartphone interaction and node signaling.

ACHIEVEMENTS FOR THE WEEK:

- Advanced the host-and-node prototype direction for addressed physical component localization.
- Refined the idea that one node can cover dense component rows while multiple nodes cover larger spaces.
- Improved the inventory sufficiency logic to include value-specific quantity checks.
- Documented the whole-project highlighting workflow for collecting all required parts together.
- Clarified the ESP32 host role as the bridge between smartphone interaction and node communication.

FUTURE WORK PLANS:

- Strengthen the inventory-aware recommendation flow on the software side.
- Continue refining the deterministic workspace and proposal-review path.
- Prepare screenshots, diagrams, and notes for formal documentation.
- Review how voice-assisted lookup could be integrated through the host bridge.

Week's Summary

Days/Time	5 Days a week Monday to Friday.
Monday	Worked on the host-and-node prototype concept for addressed component localization.
Tuesday	Refined node coverage for both dense storage rows and distributed room layouts.
Wednesday	Improved inventory logic to include value-specific quantity sufficiency checks.
Thursday	Documented the whole-project highlighting workflow for selected circuit builds.
Friday	Clarified the ESP32 host role in bridging smartphone interaction and node signaling.

STATUS OF RESEARCH PAPER: The hardware architecture and inventory-aware problem statement were refined for use in synopsis and paper drafting. Estimated overall project completion by the end of this week: 49%.

Supervisor signature: **Date:**